

Retinitis Pigmentosa 59 (RP59): A Comprehensive Disease Characteristics Report

Disease: Retinitis Pigmentosa 59 | **OMIM:** #613861 | **MONDO:** MONDO:0013468 | **Gene:** *DHDDS* (OMIM *608172) | **Category:** Mendelian (autosomal recessive)

Summary

Retinitis Pigmentosa 59 (RP59) is a rare, non-syndromic, autosomal recessive rod-cone dystrophy caused by biallelic missense variants in **DHDDS** (dehydrodolichyl diphosphate synthase), the catalytic subunit of the endoplasmic-reticulum **cis-prenyltransferase (cis-PT)** complex that synthesizes dolichol — the obligate lipid carrier for protein N-glycosylation. The disease is defined biochemically by a hypomorphic enzymatic defect: the recurrent **K42E (c.124A>G, p.Lys42Glu)** allele lowers catalytic efficiency and shortens dolichol chains (yielding a diagnostic elevation of the dolichol-18/dolichol-19 ratio) **without** producing a gross serum hypoglycosylation defect. K42E is an **Ashkenazi Jewish (AJ) founder mutation**, accounting for ~33% of genetically solved AJ retinitis pigmentosa families and carried by roughly **1 in 91 people of AJ ancestry** (gnomAD v4 ASJ allele frequency 0.55%).

Clinically, RP59 presents as a classic rod-cone dystrophy — night blindness, progressive peripheral-then-central visual-field loss, bone-spicule retinal pigmentation, and an attenuated or extinguished electroretinogram (ERG) — but with **unusually prominent macular and retinal-pigment-epithelium (RPE) involvement** that gives a distinctive fundus autofluorescence signature. This distinguishes DHDDS-RP from other genetic RP subtypes and hints at a mechanism extending beyond pure photoreceptor loss. Multiple knock-in and conditional mouse models (K42E, T206A, RPE-specific *Dhdds* ablation) localize early pathology to the **inner retina** — thinning of the inner nuclear layer, reduced bipolar/amacrine cell densities, and defective photoreceptor-to-bipolar synaptic transmission — as well as to the RPE, rather than to primary outer-segment degeneration.

There is currently **no gene-specific cure**. Management is supportive: carbonic-anhydrase inhibitors for cystoid macular edema, cataract surgery, low-vision rehabilitation, and genetic counseling with founder-variant carrier screening in Ashkenazi Jews. Antioxidant (N-acetylcysteine) and broad gene-based therapies for RP are in development. Importantly, *DHDDS* is an allelic locus for a **separate, dominant, de novo neurodevelopmental/neurodegenerative disorder (DEDSM)** — developmental delay, epilepsy, myoclonus, and movement disorder — which is mechanistically and genetically distinct from recessive RP59.

Section 1 — Disease Information

Overview. RP59 is a specific, rare, autosomal recessive, non-syndromic subtype of retinitis pigmentosa. Retinitis pigmentosa as a whole is a leading cause of inherited visual disability with a worldwide prevalence of approximately **1:4000** ([PMID: 29597005](#); "*RP is a leading cause of visual disability, with a worldwide prevalence of 1:4000.*"). RP59 is defined by biallelic pathogenic variants in *DHDDS*.

Key identifiers.

Resource	Identifier
OMIM (disease)	#613861 (Retinitis pigmentosa 59)
OMIM (gene)	*608172 (<i>DHDDS</i>)
MONDO	MONDO:0013468
HGNC	HGNC:20603 (<i>DHDDS</i>)
UniProt	Q86SQ9 (<i>DHDDS</i> protein)
Gene locus	Chromosome 1p36.11
dbSNP (K42E)	rs147394623

Synonyms / alternative names. RP59; *DHDDS*-related retinitis pigmentosa; *DHDDS*-associated inherited retinal degeneration (IRD). The broader *DHDDS* disease spectrum also includes *DHDDS*-congenital disorder of glycosylation (*DHDDS*-CDG) and DEDSM (developmental delay and seizures with or without movement abnormalities).

Source of information. The knowledge here is derived from aggregated disease-level resources (OMIM, Orphanet, gnomAD) and primary literature (case series, family-based whole-exome studies, biochemical and animal-model work), not from individual EHR records.

Section 2 — Etiology

Disease causal factors. RP59 is a **monogenic Mendelian disorder** caused by biallelic (homozygous or compound heterozygous) pathogenic missense variants in *DHDDS*. Whole-exome sequencing of an Ashkenazi Jewish family with 3 of 4 affected siblings identified a homozygous **c.124A>G (p.Lys42Glu, K42E)** variant as causal ([PMID: 24664694](#); *"A single-nucleotide mutation in the gene that encodes DHDDS has been identified by whole exome sequencing as the cause of the non-syndromic recessive retinitis pigmentosa (RP) in a family of Ashkenazi Jewish origin..."*). Reported RP59-causing genotypes include **K42E/K42E, T206A/K42E, and R98W/K42E** ([PMID: 40574710](#); *"three variant alleles (K42E/K42E, T206A/K42E and R98W/K42E) have been reported to cause retinitis pigmentosa 59 (RP59)"*).

Genetic risk factors. The single dominant genetic risk factor is inheritance of two pathogenic *DHDDS* alleles. The **K42E founder allele** dramatically elevates carrier risk in individuals of Ashkenazi Jewish ancestry (see Sections 4 and 9). Modifier genes in the glycosylation pathway (notably *ALG6*) influence expressivity (Section 4).

Environmental risk factors. No established environmental risk factors initiate RP59; the disease is fully genetically determined. General retinal-health factors (e.g., light exposure, oxidative stress) may modulate progression but are not documented as specific RP59 risk factors.

Protective factors. No validated genetic or environmental protective alleles specific to RP59 are documented. Modifier alleles can shift severity in either direction (e.g., *ALG6* F304S was associated with **less** peripheral rod disease even while worsening macular cone disease; [PMID: 38256083](#)).

Gene–environment interactions. No specific gene–environment interaction has been established for RP59. The dominant modulatory interactions are **gene–gene** (glycosylation-pathway modifiers).

Section 3 — Phenotypes

RP59 manifests as a progressive rod-cone dystrophy with added macular/RPE features. Key phenotypes, with suggested HPO terms:

Phenotype	Type	HPO term	Onset / progression	Frequency
Night blindness (nyctalopia)	Symptom	HP:0000662	Early, progressive	Characteristic/typical
Rod-cone dystrophy	Clinical sign	HP:0000510	Early-onset, progressive	Defining feature
Constriction of visual field	Clinical sign	HP:0001133	Peripheral→central, progressive	Typical
Bone-spicule retinal pigmentation	Physical manifestation	HP:0007737	Progressive	Typical
Abnormal/attenuated or absent ERG	Lab/functional	HP:0000512	Early, progressive	Typical/severe
Reduced visual acuity	Symptom/sign	HP:0007663	Progressive, worse & earlier than MAK-RP	Common
Macular degeneration / maculopathy	Clinical sign	HP:0000608	Prominent in DHDDS-RP	Characteristic
Retinal pigment epithelial atrophy	Clinical sign	HP:0007894	Progressive	Common

Characteristics. DHDDS-RP59 patients exhibit **classic RP symptoms** (night blindness, progressive peripheral then central visual field loss, bone-spicule pigmentation, attenuated ERG) **plus macular changes suggestive of RPE involvement** (PMID: 32245241; "Patients with certain defects in the dehydrololichyl diphosphate synthase (DHDDS) gene (RP59; OMIM #613861) exhibit classic symptoms of retinitis pigmentosa, as well as macular changes, suggestive of retinal pigment epithelium (RPE) involvement."). On ultra-widefield fundus autofluorescence (FAF), DHDDS patients (n=12) had a **significantly more abnormal macular FAF pattern and more widespread decreased peripheral autofluorescence** than MAK or

FAM161A RP patients (macular abnormality $p=0.001$) (PMID: 35501492; "DHDDS patients had a more abnormal macular FAF pattern and more widespread decrease in peripheral autofluorescence."). DHDDS patients tend to have **worse visual acuity and visual fields at younger ages than MAK patients** (PMID: 29276052), indicating relatively severe, earlier expressivity.

Severity / progression. Severity is **moderate-to-severe and progressive**, with variable expressivity influenced by modifier genes. Onset is early (see Section 8).

Quality-of-life impact. Progressive constriction of the visual field and central vision loss lead to loss of independent mobility, driving cessation, reading difficulty, and eventual legal blindness — substantial impacts on daily functioning. Disease-specific QoL instrument data for RP59 specifically were not identified; general RP QoL burden applies.

Section 4 — Genetic / Molecular Information

Causal gene. *DHDDS* (dehydrodolichyl diphosphate synthase), chromosome 1p36.11, OMIM *608172, HGNC:20603, UniProt Q86SQ9.

Pathogenic variants.

Variant	HGVS (NM_024887.4)	Protein	Type	Role
K42E	c.124A>G	p.Lys42Glu	Missense	Founder hypomorph; most common RP59 allele
T206A	c.616A>G	p.Thr206Ala	Missense	RP59 in trans with K42E or homozygous
R98W	—	p.Arg98Trp	Missense	RP59 in trans with K42E

The K42E variant changes the **highly conserved residue Lys42 to Glu, resulting in lower catalytic efficiency** (PMID: 24664694) — i.e., a **hypomorphic, partial loss-of-function** allele rather than a null. Biallelic null/severe-hypomorphic combinations (e.g., nonsense + splice) instead cause fatal infantile DHDDS-CDG, indicating an allelic dosage/severity spectrum.

Variant classification (ACMG/AMP). K42E is classified **pathogenic** for RP59 given segregation, functional enzyme data, and biochemical biomarker evidence. T206A and R98W are pathogenic/likely-pathogenic in the recessive RP context.

Allele frequency (gnomAD v4; computational/database evidence — Finding F012). Direct query of gnomAD v4 for variant 1-26438228-A-G (rs147394623) returns:

Population	Allele frequency	Allele count
Global (exomes)	1.29×10^{-4}	188 / 1,461,728
Ashkenazi Jewish (ASJ)	0.555%	145 / 26,132
Non-Finnish European	2.2×10^{-5}	24 / 1,111,908
African / East Asian / South Asian / Finnish / Mid-Eastern	~absent	—

This is a **~257-fold ASJ-vs-NFE enrichment**. Predicted ASJ carrier frequency $\approx 2pq \approx 1.1\%$ (**~1 in 91**); predicted ASJ homozygote (affected) frequency $q^2 \approx 1$ **in ~32,000** before accounting for compound heterozygosity with T206A/R98W.

Functional consequence. Partial **loss of function** (reduced catalytic efficiency of cis-PT), not gain-of-function or dominant-negative, in the recessive RP context.

Modifier genes. In 11 K42E-IRD patients, an **ALG6 modifier variant (F304S)** correlated with **greater macular cone disease severity but less peripheral rod disease severity**, showing that glycosylation-pathway modifier genes influence RP59 expressivity (PMID: 38256083). Additional candidate modifiers tested include *ALG8*, *DDOST*, *MPDU1*, *TNKS*.

Epigenetic information / chromosomal abnormalities. No disease-specific DNA-methylation, histone, or large-scale chromosomal abnormality is documented for RP59; it is a point-mutation monogenic disorder.

Section 5 — Environmental Information

No environmental toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents are established as causes or triggers of RP59. The disease is entirely genetically determined by biallelic *DHDDS* variants. (This section is **not applicable** beyond general retinal-health considerations.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic hypomorphic DHDDS variants (e.g., K42E)** → reduce the **catalytic efficiency** of the DHDDS subunit of cis-prenyltransferase (*demonstrated* — enzyme assays, [PMID: 24664694]).
2. Reduced cis-PT activity → **impaired synthesis of dehydrodolichyl diphosphate**, the precursor of dolichol (*demonstrated* — enzymology/structure, [PMID: 33077723]).
3. Impaired dolichol synthesis → **characteristic shortening of dolichol chains**, with dolichol-18 replacing normal dolichol-19 (elevated D18/D19 ratio) in plasma/urine/retina (*demonstrated* — LC-MS in patients and mice, [PMID: 24078709]; [PMID: 37443173]).
4. Branch A — **N-glycosylation**: In RP59, dolichol shortening does **NOT** cause gross serum hypoglycosylation (transferrin isoelectric focusing normal; [PMID: 24664694]). This distinguishes RP59 from severe DHDDS-CDG.
5. Branch B — **Retinal/RPE pathology** (the clinically relevant path): altered dolichol metabolism → **inner-retinal dysfunction** with defective photoreceptor-to-bipolar synaptic transmission, inner nuclear layer thinning, and bipolar/amacrine cell loss (*demonstrated in mouse models*, inferred for humans; [PMID: 37443173]; [PMID: 40574710]) **and** RPE atrophy/dysfunction (*demonstrated* in RPE-specific ablation mice; [PMID: 32245241]).
6. Inner-retinal + RPE dysfunction → **progressive rod-cone degeneration, macular/RPE involvement, and abnormal ERG/FAF** → clinical **night blindness, field constriction, central vision loss** (*clinical manifestation*).

Detail by category

Molecular pathways / biochemistry. DHDDS is the catalytic subunit of the human cis-prenyltransferase (with NgBR/NUS1), synthesizing **dehydrodolichyl diphosphate**, the precursor of **dolichol**, the obligate lipid carrier for N-glycosylation (PMID: 33077723; "The human cis-prenyltransferase (hcis-PT) is an enzymatic complex essential for protein N-glycosylation. Synthesizing the precursor of the glycosyl carrier dolichol-phosphate, mutations in hcis-PT cause severe human diseases."). The enzyme operates in the isoprenoid/dolichol arm of the mevalonate pathway. GO terms: **di-trans,poly-cis-decaprenylcistransferase activity** (GO:0045547), **dolichol biosynthetic process** (GO:0019408), **ER membrane** (GO:0005789).

Protein dysfunction / structure. The human cis-PT is a **heterotetramer** of two catalytic DHDDS subunits and two inactive **Nogo-B receptor (NgBR, gene NUS1)** subunits, assembling via DHDDS C-termini as a **dimer-of-heterodimers**; the NgBR distal C-terminus crosses the interface to help form the DHDDS active site (2.3 Å crystal structure; [PMID: 33077723](#); *"the 2.3 Å crystal structure reveals that the tetramer assembles via the DHDDS C-termini as a dimer-of-heterodimers"*). **Disease mutations cluster around the active site**, and molecular-dynamics simulations propose a mechanism for hcis-PT dysfunction in RP ([PMID: 33077723](#); *"we explored the functional consequences of disease mutations clustered around the active-site... we propose a mechanism for hcis-PT dysfunction in retinitis pigmentosa"*). K42 is a highly conserved residue; K42E is a hypomorphic partial loss-of-function change.

Cellular processes / tissue damage. Mouse models show **elevated expression of synaptogenesis/synaptic genes**, progressive reduction of inner nuclear layer (INL) and total retinal thickness from ~postnatal 2 months, and INL/outer plexiform layer cell loss — **without profound photoreceptor outer-segment degeneration or N-glycosylation defect** ([PMID: 37443173](#); *"Quantitative retinal cell layer thickness measurements demonstrated a significant reduction in the inner nuclear layer (INL) and total retinal thickness (TRT) beginning at postnatal (PN) ~2 months"*). T206A/T206A, T206A/K42E and K42E/K42E mice show reduced INL thickness, reduced ERG b-waves with relatively spared a-waves, attenuated c- and d-waves, and reduced bipolar/amacrine densities. The authors **propose that RP59 dysfunction involves defective photoreceptor-to-bipolar synaptic transmission with concomitant bipolar/amacrine cell degeneration** ([PMID: 40574710](#); *"We propose that the physiological basis of retinal dysfunction in RP59 involves defective photoreceptor to bipolar cell synaptic transmission with concomitant bipolar/amacrine cell degeneration."*).

RPE contribution. Selective *Dhdds* ablation in mouse RPE causes **RPE atrophy, hyper-reflectivity, transmigration into the photoreceptor layer, and scotopic a-/b-wave reductions of 83%/77% at 3 months** ([PMID: 32245241](#)), directly demonstrating an RPE-autonomous component consistent with the macular/RPE phenotype seen in patients.

Cell types (CL) and biological processes (GO). Cell types: photoreceptor (CL:0000210), rod (CL:0000604), cone (CL:0000573), retinal bipolar neuron (CL:0000748), amacrine cell (CL:0000561), retinal pigment epithelial cell (CL:0002586). Biological processes: dolichol biosynthetic process (GO:0019408), protein N-linked glycosylation (GO:0006487), synaptic transmission / photoreceptor cell maintenance.

Immune / metabolic / epigenetic. No autoimmune or infectious mechanism. The core metabolic defect is in **isoprenoid/dolichol lipid metabolism**. No RP59-specific epigenetic mechanism is documented.

Section 7 — Anatomical Structures Affected

- **Organ level:** Eye — specifically the **retina** (UBERON:0000966) and **retinal pigment epithelium** (UBERON:0001782). Body system: **visual/nervous system**. RP59 is non-syndromic — no secondary organ involvement in the classic recessive form (contrast: DHDDS-CDG and DEDSM involve the CNS).
- **Tissue/cell level:** Neural retina (nervous tissue) and RPE (epithelial tissue). Affected cell populations: rods, cones, **retinal bipolar neurons (CL:0000748)** and **amacrine cells (CL:0000561)** (inner retina), and **RPE cells (CL:0002586)**. The inner-nuclear-layer emphasis is a distinctive feature relative to classic outer-retinal RP.
- **Subcellular level:** **Endoplasmic reticulum membrane (GO:0005789)**, where cis-PT resides and dolichol is synthesized; photoreceptor outer segments and synaptic terminals (outer plexiform layer) are functionally implicated.
- **Localization:** **Bilateral**, roughly symmetric retinal involvement, with prominent **macular** (central) plus peripheral involvement. UBERON: retina (UBERON:0000966), macula lutea (UBERON:0005388), RPE (UBERON:0001782).

Section 8 — Temporal Development

- **Onset:** Early-onset retinal degeneration; the index AJ family had **early-onset** disease in 3 of 4 affected siblings ([PMID: 24664694]). RP typically manifests with **night blindness in adolescence**, followed by concentric visual field loss ([PMID: 29597005](#)). DHDDS patients tend to have worse acuity/fields at **younger ages** than MAK patients ([PMID: 29276052]).
- **Onset pattern:** Insidious, chronic.
- **Progression:** Progressive and centripetal — *"RP typically manifests with night blindness in adolescence, followed by concentric visual field loss, reflecting the principal dysfunction of rod photoreceptors; central vision loss occurs later in life due to cone dysfunction"* ([PMID: 29597005](#)). Stages: early (nyctalopia, mid-peripheral field loss) → intermediate (ring scotoma, tunnel vision) → advanced (central vision loss, near-extinguished ERG) → end-stage (legal blindness).
- **Course / duration:** Chronic, lifelong, progressive.

- **Remission / critical periods:** No spontaneous remission. Early intervention (before substantial cell loss) is the theoretical window for future gene/cell therapies; mouse INL thinning begins ~PN 2 months, defining an early structural window.

Section 9 — Inheritance and Population

- **Epidemiology:** RP overall prevalence ~**1:4000** ([PMID: 29597005]). RP59 is a specific **rare** autosomal recessive subtype; precise population incidence/prevalence of RP59 is not separately established, but it is enriched in Ashkenazi Jews. Among 230 AJ RP families, a cause was found in **37%**, and K42E was present in **33%** of solved families (second only to the MAK Alu insertion at 39%) ([PMID: 29276052](#); "...c.124A>G, p.K42E in dehydrodolichol diphosphate synthase (DHDDS) (33%).").
- **Inheritance: Autosomal recessive** with essentially complete penetrance in biallelic carriers.
- **Penetrance / expressivity:** High/complete penetrance; **variable expressivity** modulated by modifiers (e.g., *ALG6*; [PMID: 38256083]).
- **Founder effect:** K42E is an **Ashkenazi Jewish founder allele** (gnomAD ASJ AF 0.555%; ~257-fold over NFE; Finding F012). Carrier frequency ~**1 in 91** in AJ ancestry.
- **Consanguinity:** Founder homozygosity (rather than classical consanguinity) underlies most AJ cases; consanguinity contributes in recessive RP broadly.
- **Sex ratio:** No sex predilection (autosomal).
- **Geographic/ethnic distribution:** Concentrated in Ashkenazi Jewish populations; the variant is essentially absent in African, East Asian, South Asian, Finnish, and Mid-Eastern subsets (gnomAD v4).

Section 10 — Diagnostics

Diagnosis rests on **three complementary pillars** (Finding F011):

1. **Clinical RP evaluation** — fundus exam (bone-spicule pigment, attenuated vessels, waxy disc pallor); **full-field ERG** (reduced/absent scotopic and photopic responses); visual fields; **OCT** (outer retinal layer loss, macular changes); and **fundus autofluorescence** showing a

distinctive abnormal macular pattern and widespread peripheral decreased autofluorescence (PMID: 35501492).

2. **Biochemical biomarker — dolichol profiling.** LC-MS of plasma/urine shows a **characteristic shortening of dolichols** with an elevated **D18/D19 ratio** that discriminates patients > carriers > normals by ROC analysis (PMID: 24078709; *"We observed a characteristic shortening of plasma and urinary dolichols in retinitis pigmentosa (RP) patients carrying K42E and T206A mutations..."*). Crucially, *"Dolichol profiling, complementary to genotyping, can be readily adapted as a test in the clinic not only for the diagnosis of patients but also for identification of carriers with DHDDS or other genetic mutations that may impair dolichol biosynthesis."* (PMID: 24078709).
3. **Molecular genetic testing** — targeted single-variant testing for the AJ founder **K42E**, RP/IRD gene panels, or whole-exome/genome sequencing (PMID: 24664694).

Important negative test: Standard serum **transferrin isoelectric focusing is typically normal** in RP59 — *"Patterns of plasma transferrin isoelectric focusing gel were normal in all family members, indicating no significant abnormality in protein glycosylation"* (PMID: 24664694) — so routine CDG screening will miss RP59; dolichol profiling and/or genetics are required.

Differential diagnosis: other genetic RP subtypes (MAK, FAM161A, USH2A, RPGR), Leber congenital amaurosis, and acquired outer retinopathies (e.g., AZOOR) — distinguished by genotype and the distinctive DHDDS FAF/macular pattern.

Screening: Ashkenazi Jewish **carrier screening** for K42E (and cascade family testing) is the key preventive-diagnostic measure.

CHEBI terms: dolichol (CHEBI:16091), dolichyl phosphate (CHEBI:57683).

Section 11 — Outcome / Prognosis

- **Survival/mortality:** RP59 is **not life-limiting** in its non-syndromic recessive form; no excess mortality. (Contrast: severe biallelic DHDDS-CDG can be fatal in infancy.)
- **Morbidity/function:** Progressive, potentially severe **visual disability** culminating in legal blindness; DHDDS patients trend toward worse acuity/fields at younger ages than MAK-RP (PMID: 29276052).

- **Disease course/complications:** Cystoid macular edema, cataract, and posterior subcapsular lens changes are common RP complications affecting central vision.
 - **Prognostic factors:** Genotype (K42E/K42E vs compound heterozygous), modifier alleles (*ALG6* F304S shifts macular vs peripheral severity; [PMID: 38256083]), and age at presentation.
 - **Prognostic biomarker:** Dolichol D18/D19 ratio tracks the biochemical defect and could serve as a monitoring/prognostic marker (research use).
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Section 12 — Treatment

There is no approved gene-specific therapy for DHDDS-RP59 (Finding F009). Management is supportive and symptomatic.

Intervention	Evidence / role	NCIT (suggested)
Carbonic-anhydrase inhibitors (oral acetazolamide, methazolamide; topical dorzolamide)	First-line for RP-associated cystoid macular edema. Network meta-analysis (12 studies) found these reduced central macular thickness and improved BCVA at 3–4 months, outperforming anti-VEGF/steroids (PMID: 42493417 ; "At 3-4 months, DEXi, acetazolamide, methazolamide and dorzolamide demonstrated comparable CMT reduction and outperformed anti-VEGF therapies")	Acetazolamide (NCIT:C233); Dorzolamide
N-acetylcysteine (NAC)	Antioxidant; Phase III " NAC Attack " trial ongoing for RP, targeting oxidative-stress-driven photoreceptor loss (PMID: 39864434 ; "The ongoing multicentre Phase III trial 'NAC Attack' aims to evaluate the long-term efficacy and safety of NAC in RP.")	N-Acetylcysteine (NCIT:C61796)
Low-vision rehabilitation, cataract surgery	Standard supportive care	Low Vision Aid
Gene / cell-based therapy	Investigational for RP broadly; none DHDDS-specific yet	Gene Therapy (NCIT:C15262)
Vitamin A palmitate	Debated in RP generally; contraindicated in some genotypes — use with caution	—

Of note, acetazolamide improved systemic symptoms in a K42E DHDDS-CDG patient ([PMID: 36046393](#)). **Personalized approach:** genotype-guided counseling and modifier-aware prognostication (*ALG6*) are emerging.

Section 13 — Prevention

- **Primary prevention:** Not applicable to a Mendelian disorder in a born individual; **genetic counseling and Ashkenazi Jewish carrier screening** for K42E enable informed reproductive decisions (prenatal testing, preimplantation genetic diagnosis). Given ~1-in-91 AJ carrier frequency, population carrier screening is impactful.

- **Secondary prevention:** Early diagnosis (genetics + dolichol profiling + imaging) to monitor and treat complications (macular edema, cataract) before irreversible central vision loss.
 - **Tertiary prevention:** Management of complications and low-vision rehabilitation to preserve function.
 - **Counseling:** Autosomal-recessive risk counseling; cascade testing of relatives of probands.
 - **Immunization / public health / environmental:** Not applicable.
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Section 14 — Other Species / Natural Disease

- **Taxonomy/orthologs:** *DHDDS* is highly conserved across eukaryotes; the enzyme's homology across species enables cross-species mechanistic study ([PMID: 28809830](#)). Mouse *Dhdds* is the principal ortholog used for modeling.
 - **Natural disease in other species:** No well-characterized spontaneous RP59-equivalent natural disease is documented in companion animals or wildlife; vertebrate models are engineered (see Section 15; reviewed in [PMID: 36362109](#)).
 - **Comparative biology:** The cis-PT/dolichol pathway is evolutionarily conserved from bacteria/yeast (rubber/undecaprenyl synthases) to humans, supporting strong conservation of disease mechanism.
 - **Transmission / zoonosis:** Not applicable (genetic disease).
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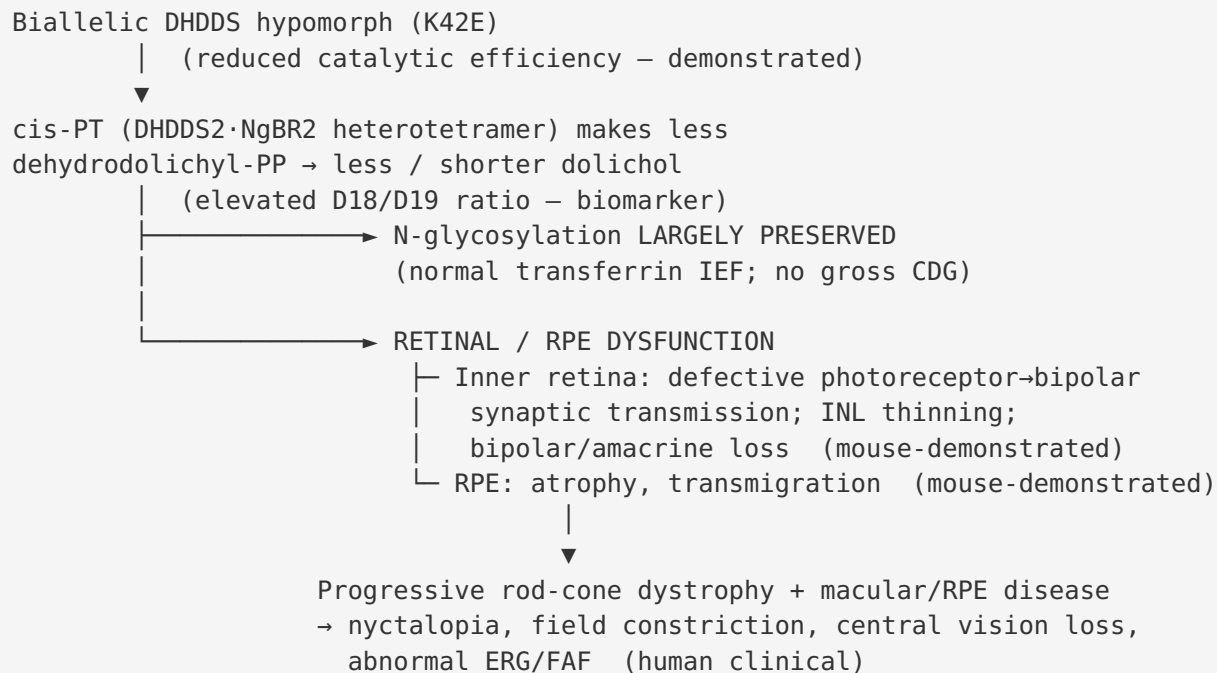
Section 15 — Model Organisms

RP59 has a rich set of engineered models (reviewed in *Vertebrate Animal Models of RP59*, [PMID: 36362109](#)).

Model	Type	Key phenotype	Reference
Dhdds K42E knock-in mouse	Mammalian, knock-in	Shortened retina/liver/brain dolichols; INL & total retinal thinning from ~PN 2 mo; INL/OPL cell loss; no profound outer-segment degeneration or N-glycosylation defect; defective synaptic transmission	PMID: 37443173
Dhdds T206A & K42E knock-in mice (T206A/T206A, T206A/K42E, K42E/K42E)	Mammalian, knock-in	Reduced INL thickness; reduced ERG b-waves with relatively spared a-waves; attenuated c-/d-waves; reduced bipolar/amacrine densities; phenotypically similar across genotypes	PMID: 40574710
RPE-specific Dhdds ablation mouse	Mammalian, conditional KO	RPE atrophy, hyper-reflectivity, transmigration into photoreceptor layer; scotopic a-/b-wave reductions 83%/77% at 3 mo	PMID: 32245241
Patient-derived cells	In vitro	Niemann-Pick C-like endolysosomal dysfunction; correctable by miglustat (in DHDDS-CDG context)	PMID: 40003936

Phenotype recapitulation: The knock-in models reproduce the **inner-retinal** and RPE features but, notably, do **not** show the profound photoreceptor degeneration classically expected of RP — a key insight redirecting the mechanistic focus to bipolar/amacrine synaptic pathology. **Limitations:** species differences in dolichol chain length and retinal architecture; incomplete modeling of human macular involvement (mice lack a macula).

Mechanistic Model / Interpretation



The central, somewhat counterintuitive insight of RP59 biology is a **tissue-specific vulnerability paradox**: DHDDS is essential for N-glycosylation in *all* cells, yet the recessive hypomorphic K42E allele produces an **eye-restricted phenotype without gross systemic hypoglycosylation**. The retina — and specifically the inner-retinal synaptic circuitry and RPE — appears exquisitely sensitive to the partial reduction in dolichol supply. This contrasts with the **allelic dominant DEDSM** disorder (de novo *DHDDS* variants such as R211Q, R37H) that causes a neurodevelopmental/neurodegenerative syndrome with epilepsy, myoclonus, and movement disorder ([PMID: 34382076](#); "Patients presented during infancy or childhood with a variable association of neurodevelopmental disorder, generalized epilepsy, action myoclonus/cortical tremor and ataxia."), and with severe biallelic null combinations that cause fatal infantile CDG. RP59 thus sits at the mild end of a *DHDDS* allelic dosage spectrum.

Evidence Base

PMID	Contribution	Supports
24664694	WES identifies K42E as cause of recessive RP; reduced catalytic efficiency; normal transferrin IEF	F001, F003, F007, F011
40574710	RP59-causing genotypes; knock-in mice; inner-retinal synaptic mechanism	F001, F004
24078709	Dolichol chain shortening (D18/D19) as biomarker & carrier test	F003, F011
33077723	2.3 Å cis-PT heterotetramer structure; mutations at active site	F003, F007
29276052	K42E = 33% of solved AJ RP; worse phenotype than MAK	F002, F006, F010
32245241	Macular/RPE involvement; RPE-ablation model (83%/77% ERG loss)	F006
35501492	Distinctive DHDDS FAF/macular signature	F006, F011
37443173	K42E knock-in: INL thinning, synaptic defect, no CDG	F003, F004
38256083	ALG6 F304S modifier of retinal severity	F008
34382076	De novo dominant DHDDS neurodevelopmental disorder (DEDSM)	F005, F008
40003936	DHDDS-NgBR dolichol role; endolysosomal dysfunction; miglustat	F005
36046393	Late-onset CDG from homozygous K42E; acetazolamide benefit	F008, F009
42493417	Carbonic-anhydrase inhibitors first-line for RP macular edema	F009
39864434	NAC Phase III "NAC Attack" antioxidant trial for RP	F009
29597005	RP prevalence 1:4000; natural history	F010
36362109	Review of RP59 vertebrate animal models	Section 15
gnomAD v4 (computational)	K42E ASJ AF 0.555%; ~1 in 91 carriers; 257-fold enrichment	F012

Limitations and Knowledge Gaps

1. **Tissue-specificity mechanism unresolved.** Why a ubiquitously required glycosylation enzyme causes an *eye-restricted*, largely non-glycosylation phenotype in RP59 is not mechanistically explained at the molecular level.
2. **Human inner-retinal pathology is inferred from mouse models.** The bipolar/amacrine synaptic mechanism ([PMID: 37443173]; [PMID: 40574710]) is demonstrated in mice; direct human histopathology confirmation is limited, and mice lack a macula (the very region most distinctively affected in patients).
3. **No RP59-specific prevalence/incidence figures.** Epidemiology is extrapolated from AJ RP cohorts and gnomAD carrier frequencies; true population penetrance and prevalence (including compound heterozygotes) are estimates.
4. **Limited genotype–phenotype data for non-K42E genotypes** (T206A, R98W) in humans.
5. **No dedicated QoL/natural-history longitudinal data** for RP59 specifically.
6. **Modifier landscape incomplete.** Only *ALG6* is validated; *ALG8*, *DDOST*, *MPDU1*, *TNKS* remain candidate modifiers.
7. **No disease-specific therapy** exists; the therapeutic pipeline (gene therapy, antioxidants) is generic to RP, not DHDDS-targeted.

Proposed Follow-up Experiments / Actions

1. **Human retinal validation:** Use patient-derived retinal organoids/iPSC-RPE and, where available, donor retinal histopathology to confirm the inner-retinal/bipolar-synaptic mechanism and the RPE contribution in human tissue.
2. **Dolichol-supplementation / substrate-rescue trials:** Test whether dolichol or dolichyl-phosphate supplementation, or isoprenoid-pathway modulation, rescues the biochemical and functional phenotype in K42E knock-in mice and organoids.
3. **Gene-replacement therapy development:** AAV-DHDDS gene augmentation (targeting photoreceptors/bipolar cells and RPE) given the recessive loss-of-function mechanism and early therapeutic window (pre-PN-2-month structural change in mice).

4. **Prospective natural-history study** of genotyped RP59 patients with standardized OCT, FAF, ERG, visual fields, and QoL instruments, stratified by genotype and *ALG6* modifier status.
5. **Systematic modifier screen** (*ALG6*, *ALG8*, *DDOST*, *MPDU1*, *TNKS* and genome-wide) in larger K42E cohorts to build a severity-prediction model.
6. **Biomarker qualification:** Validate plasma/urinary D18/D19 dolichol ratio as a surrogate endpoint for therapeutic trials and as a clinical carrier-screening adjunct.
7. **Expand Ashkenazi Jewish carrier screening panels** to include DHDDS K42E (carrier ~1 in 91) alongside other AJ founder mutations, with cascade testing.

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